

Adversarial correctness audit

Exact Derandomization of Logarithmic Space: $L = RL = BPL$

Subject: the 133-page deslop rewrite, compared with the 108-page original.

Standard: the supplied `verifier(4).md`. **Audit date:** 9 October 2026.

1 Verdict

VERIFIED SUBJECT TO EXPLICIT ASSUMPTIONS.

The mathematical argument checks out under the imported results listed in Section 6 and the three explicit notation identifications in Section 3. I found no fatal error, no missing load-bearing proof step, and no omitted formal result or substantive weakening of the original’s assumptions or conclusions. This verdict applies to the argument with those identified symbols interpreted as specified; it does not certify the literal well-formedness of every printed formula. In particular, the rewrite’s principal estimator inequality has an undefined summation domain. The verification is a mathematical source audit, including the probability law and simultaneous memory accounting, not an executed implementation or a proof-assistant certificate. Historical and comparative citations not needed for the derivation are explicitly outside this conditional certification.

2 Claim audited

High-level purpose. The resource model and order of quantifiers determine what is being claimed. The audit must distinguish a single algorithm with an input precision request from a different algorithm for each fixed accuracy.

Fix a randomized machine $\mathcal{M}$ with finite control, finitely many work tapes, endmarked read-only input, and fresh independent fair bits. Its input heads remain between the endmarkers. On every input of length n and every random tape it halts in polynomial time and traverses $O(\log(n+2))$ writable bits. Traversing a blank work cell counts. All simultaneously live counters, arithmetic words, and suspended records count. Output, when present, is one-way and write-only.

The main claims checked are the following.

1. **Fixed accuracy.** For each fixed integer $d \geq 1$, one deterministic uniform transducer outputs $z_x \in [0, 1]$ with

$$|z_x - p_{\mathcal{M}}(x)| \leq (n+2)^{-d}.$$

Space and output length are $O(\log(n+2))$; time is polynomial. Constants may depend on $\mathcal{M}, d$, never on x .

2. **Input accuracy.** For this fixed $\mathcal{M}$, one algorithm on $(x, 1^q)$, $q \geq 1$, outputs an integer $0 \leq a \leq 2^{q+2}$ with

$$|a2^{-(q+2)} - p_{\mathcal{M}}(x)| \leq 2^{-q}, \quad S = O(\log(n+2) + q), \quad T = (n+2)^{O(1)}2^{O(q)}.$$

The constants depend on $\mathcal{M}$, not q . The precision encoding is unary.

3. **Decision and promise consequences.** $L = RL = BPL$ with the stated one-sided and two-sided error thresholds. Promise separation uses a total decider even outside the promise.
4. **Search.** For fixed $c \geq 0$, acceptance probability at least $(n+2)^{-c}$ guarantees an accepting run, or its certified witness, in deterministic logarithmic space and polynomial time. Otherwise the output is an accepting run or failure, never an unverified run.
5. **Effectivity.** A terminating compiler takes a finite source description and supplied positive integer exponents a, b , and emits a deterministic decider and explicit space/time bounds. Correctness for the source language is conditional on the supplied worst-case resource bounds and bounded-error promise. The compiler does not decide whether those promises are true.

The proof is not a claim about unrestricted-time probabilistic logspace, binary-encoded precision with polynomial running time in its encoding length, or arbitrary writable output memory. Those extensions are absent from both documents.

Pinned material and page convention

R denotes the rewrite; **O** denotes the original. Page numbers below are PDF pages, which coincide with their printed numbers. The two URLs use moving branches; the audit used these immutable revisions:

- **R**: repository `sragavan99/deslop`, commit `1925464192f95f22ca77abefdb412ca01e3e42b2`, file `bp1_example/current/main.pdf`; 133 pages, 1,275,663 bytes.
- **O**: repository `openai/math`, commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`, file `preprints/Exact-Derandomization-of-Logarithmic-Space-L-equals-RL-equals-BPL-September-23-2026/paper.pdf`; 108 pages, 1,000,094 bytes.

The rewrite cites original commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Its build source is identical to the audited original’s build source: the root TeX, preamble, bibliography, figures tree, and sections tree have the same Git object identifiers. This is a source comparison; it is not an assertion that two separately generated PDF binaries must match.

3 Load-bearing findings

Fatal error: none identified. Load-bearing gap: none identified. Local gap: none identified. The following are concrete **Presentation issues**. They are not silently repaired in the verdict.

F1. Wrong domain in the principal estimator theorem

High-level purpose. The moment estimate must range over stage vertices. An undefined or mistyped domain would make the theorem unusable literally, even if its proof establishes the intended estimate.

R, Theorem 11.10, p. 79, Eq. (11.14), source line 5002, prints

$$\frac{1}{n_v} \mathbb{E}_\sigma \sum_{x \in \mathbf{v}_l} \frac{X_u(x)}{u} |\bar{\mathbf{w}}_{l,x} - \mathbf{w}_{l,x}|^s \leq c_{\text{ind}}^s 2^{-s p_{\text{acc}}}.$$

The domain $\mathbf{v}_l$ is not defined as a set. The stage domain is V_l , defined in Section 5; $\mathbf{v}$ is also used elsewhere for vectors, including the catalyst. **O**, Theorem 9.8, p. 61, uses $x \in V_l$. The necessary correction is exactly

$$\sum_{x \in \mathbf{v}_l} \rightarrow \sum_{x \in V_l}.$$

The ordinary vector norm, the coupling proof, and Lemma 14.3 all use the vertex-domain interpretation. I checked the corrected inequality and its later use with this domain. This finding does not supply a counterexample to that corrected inequality.

F2. Stale detector-rank symbol

High-level purpose. The detector order statistic must use the same fixed rank in its definition, implementation, and error comparison. A second unnamed rank could change the construction.

R, Section 7.4, p. 37, immediately after Eq. (7.16), source line 2054, calls the column trial the J th largest score. The rewrite renamed the detector rank to j_{det} , including Eq. (5.8). Read this occurrence as j_{det} th. **O** used J consistently here. There is no indication of a new rank parameter; the same sentence explicitly identifies the trial with the earlier detector.

F3. Stale row-gate symbol

High-level purpose. The equality between the atom’s row gate and the pilot’s gate is used to identify the detector trial. It needs the actual earlier object, not an undeclared name.

R, p. 33, source line 1812, identifies the gate with R_i in Eq. (5.7). That equation instead defines $\mathbf{r}_{i,x}^{\text{gate}}$. Replace the stale R_i by that row-gate value, with the same row, sweep, and retained-column arguments. The equality follows from the explicitly matched cutoffs $\rho(t) = R_{\delta_0}(t)$; this identification was checked. **O** uses R_i for its pilot gate.

One further awkward renaming is harmless as defined: **R**, p. 74, writes $\mathcal{P}_{l_{\text{max}}}^{\text{ac}}$ for the *left* exposed-port relation, where **O** uses $\mathcal{P}_L^{\text{act}}$. The rewrite locally defines the relation, so its misleading subscript does not change the proof. A subscript such as “left” would avoid confusion with hierarchy depth.

4 Detailed verification

4.1 Dependency order and comparison scope

High-level purpose. Agreement with the original is not evidence of correctness. This ordering records the dependencies checked independently, then separates that check from preservation of the original’s claims.

The dependency chain is: configuration model $\rightarrow$ rank channels $\rightarrow$ exact hierarchy and atom interfaces $\rightarrow$ conditional averaging $\rightarrow$ prefix estimates $\rightarrow$ compression $\rightarrow$ coupled fingerprint/error induction $\rightarrow$ arithmetic and controller realization $\rightarrow$ designated-start median $\rightarrow$ decision, precision, search, and compiler consequences. Arithmetic can be proved separately before the controller despite its proofs appearing later in the rewrite. The controller is constructed by a decreasing allowance, not by assuming the estimator is already an oracle. The statistical induction does not assume accurate estimates on every environment.

For compact formulas in this report, $B = b_{\text{mem}}$ and $N = n_v$ are the rewrite’s size and stage-size bounds. These are aliases, not extra parameters. “Verified” below means the local mathematical argument was checked; “conditionally verified” indicates dependence on the imported statements or the explicit F1–F3 readings.

4.2 Configuration reduction and sampling

High-level purpose. The algorithm needs a finite forward system at every configuration, and a short random description whose conditioning remains controlled. Failure here would invalidate both the hierarchy and the final enumeration.

Status: verified.

R Lemmas 3.1 and 4.1 (**O** 2.1 and 3.1).

The time coordinate makes the transition matrix $\mathbf{S}$ nilpotent. Retaining each halted configuration through subsequent layers, and placing acceptance only in the final layer, prevents multiple reward collection. Thus

$$(\mathbf{I} - \mathbf{S})^{-1} \mathbf{e} = \sum_{j \geq 0} \mathbf{S}^j \mathbf{e} \in [0, 1]^{V_0}$$

is a finite sum even at valid but unreachable configurations. Constant indegree follows by guessing the finitely many predecessor control states, overwritten symbols, head moves, and coin choices. Duplicate predecessor descriptions are merged locally. Missing mass from invalid transitions is compatible with substochasticity.

For a sampling channel, the probability of a full-rank frame is

$$q_{\text{fr}} = \prod_{j=0}^{h_0-2} (1 - p_{\text{fld}}^{j-h}) \geq c_{\text{fr}} > 0.$$

The independent uniform constant column makes a single evaluation uniform even after frame conditioning. With $l_k = \lceil p_{\text{fld}} 2^{-k} \rceil$ and $p_{\text{fld}} 2^{-k} \geq 1$, its retention probability lies between 2^{-hk} and $2^h 2^{-hk}$. Before conditioning, distinct evaluations are pairwise independent because the evaluation functionals have rank two. Conditioning costs at most q_{fr}^{-1} , and division by the unchanged start marginal gives the stated conditional pair bound. Chebyshev gives the hit-count lower-tail bound $4/(q\pi_a q_{\text{fr}})$ for a deterministic set of $q > 0$ identifiers. No mutual independence of all hits is used.

An affine identifier substitution has an invertible triangular action on the nonconstant coefficient columns. It preserves frame rank and uniform measure. Injective identifiers remain distinct modulo the chosen prime; this is necessary and is paid for when the prime is chosen.

4.3 Exact hierarchy, detector, and copying

High-level purpose. A sparse estimate is useful only if its exact target approaches the original probability. This block checks progress, positivity, and the cost of exceptional columns.

Status: verified.

R Lemmas 5.2–5.5 and Propositions 5.6, 5.8 (**O** 4.1–4.5, 4.7).

For fixed $0 \leq \mathbf{g} \leq 1$, the fixed-point equation defines each row successively in increasing column order. Its residual

$$\mathbf{T} = \mathbf{C} - \mathbf{E}_* + \mathbf{E}_* \mathbf{C}$$

is nonnegative, and $\mathbf{T}\mathbf{1} \leq \mathbf{1}$. Every positive correction leaves at least $\theta_1 f$ residual in that position. Hence the corrected support of a *single row* has size at most $1/(\theta_1 f)$. Unrolling only along that row's increasing support gives

$$\mathbf{E}_{*,xy} \leq f(1+f)^{1/(\theta_1 f)} \leq f e^{1/\theta_1}, \quad \frac{(\mathbf{E}_* - \mathbf{E}_k)_{xy}}{f} \leq e^{1/\theta_1} \frac{(L_Q/\theta_1)^k}{k!}.$$

The intermediate vertices are always correction entries with the same source x ; replacing that fact by an arbitrary graph-path count would be invalid. The stated constant sweep count is therefore justified independently of graph size.

Pilot row masses and entry bounds grow at most linearly in the fixed sweep count. The running maximum makes the pilot multipliers nonincreasing; the frozen multiplier is below each pilot multiplier. Induction gives $\mathbf{E}_i \leq \hat{\mathbf{E}}_i$, while the restarted actual sequence is monotone. The detector proof combines a conditional row-gate failure bound with Markov and the hit-count lower tail. Its two exceptional probabilities are at most $1/40$ each, yielding $19/20$. Conversely, a trial expectation greater than $1/4$ forces more than $j_{\text{det}}/(4c_\pi^+ f)$ substantial predecessors. Neither direction requires independent gate failures.

Writing the last raw input as $\mathbf{A}$ and the nonnegative increment as $(\mathbf{E}_k - \mathbf{E}_{k-1})\mathbf{C}$ proves

$$\mathbf{D} = \mathbf{C} - \mathbf{E} + \mathbf{E}\mathbf{C} \geq 0, \quad \mathbf{D}\mathbf{1} \leq \mathbf{1}, \quad \mathbf{D} \geq \frac{\theta_1}{2k_*}(\mathbf{C} + \mathbf{E} + \mathbf{E}\mathbf{C}).$$

Thus subtraction is controlled by the exact nonnegative target; it is not assumed that estimated raw arrays remain nonnegative. Fully enabled columns have the claimed small residual by the factorial truncation bound.

The routing fractions sum to one. Whenever an extra destination copy receives positive incoming mass, its row factor is one: the residual exceeds the routing threshold, so the destination cannot be a fully enabled good column. This proves $\mathbf{C}^+\mathbf{P}^r = \mathbf{P}^r\mathbf{D}$, rather than merely a row-sum identity. The detector incidence count pays for all extra active copies. The fixed choice $d_0 > k_{\text{entry}}2^h$ gives entry bound $f/2^h$; active count grows by at most two.

Inductively,

$$\mathbf{C}_l\mathbf{P}_l^c = \mathbf{P}_l^c - \mathbf{B}_l(\mathbf{I} - \mathbf{S}), \quad \mathbf{w}_l = \mathbf{B}_l\mathbf{e} = \mathbf{P}_l^c\mathbf{p}_0 - \mathbf{C}_l\mathbf{P}_l^c\mathbf{p}_0.$$

Nonnegative updates give $0 \leq \mathbf{w}_l \leq \mathbf{1}$. At the primary copy of x the error is at most $|V_0|2^{-(h-1)l}$. This uses the active-column count as well as the entry bound. Full stage domains can be larger, but still have $O(B)$ -bit addresses.

4.4 Atom normalization, support margins, and access

High-level purpose. The algorithm must compensate for sampling and gate bias while retaining valid access on inaccurate arrays. This is also where errors at a midpoint acquire an incoming-degree cost.

Status: verified, with F2–F3 explicitly identified.

R Lemmas 7.2–7.8 (**O** 5.1–5.3, 5.5–5.7).

The magnitude partition telescopes and is nonnegative. Dense and correction tables divide by preparatory conditional gate expectations; they do not divide merely by the sampling rate. The lower bounds are needed only where the numerators can be nonzero. For corrections with multiplier zero, both target and output are zero even if the preparatory expectation vanishes, because the implemented denominator is clipped to a positive interval. The proof does not illegitimately divide by a zero expectation.

The threshold ladders have strict gaps between support, access, witness, and gate closure. Their eighth-root Lipschitz bounds follow from the order-eight vanishing of the smooth switches. They apply to arbitrary real raw estimates, including negative entries. Unlimited default copies participate in ranks but not in endpoint integration. If a default could reach an entry-support threshold, its repeated rank copies have already closed the corresponding gate; no default-only positive port is exposed.

The incoming cap applies to the entire exposed correction relation, including entries with zero weight. It deletes *all* entries of an overfull column, rather than selecting an arbitrary first k_c . The auxiliary witness is one on every exposed port, so an escaping zero-weight port remains chargeable later. Tables contain at most one entry per endpoint pair. Only correction access, not every auxiliary witness support, needs bounded incoming degree.

In assembly, differences of ranks are bounded by sup differences; summing over the union of the two final supports costs bounded incoming multiplicity. Releasing an old source changes the normalization by at most f/u in the s th power. This factor is retained in the estimator's precision calculation. Independence is used for true atom identities on independent role channels, not for products of estimated factors.

4.5 Conditional averaging and validated variants

High-level purpose. Uniform mixing alone is insufficient: a recursive call must describe its averaging operation without keeping its conditioning vertices. The two guarantees are checked separately.

Status: conditionally verified on the cited group and interpolation inputs.

R Lemmas 8.1, 8.2, 8.4 (**O** 6.1–6.3).

The two-coordinate translation projections commute. Their product projects onto invariance under all translations; the linear-group estimate then controls the remaining orthogonal component. This yields a constant distance-to-invariants bound. The lazy symmetric walk converts it to a positive spectral gap by its Dirichlet form. The constant depends on fixed dimension, not the finite action.

On a conditioned channel, the frame and retained evaluation are independent uniform coordinates. The action on $(\mathbf{F}, \mathbf{w})$ uses reduction modulo p_{fld} on the frame and modulo l_{box} on the box. It is transitive: because the frame has fewer than h columns, a complementary column can correct the determinant of a change of basis; translations then move the box coordinate. The moduli need not match. All free salt bits are concatenated into one torus component, so the number of components is bounded independently of the number of salt blocks. Interpolation gives

$$\|\mathcal{K}^d - \Pi\|_{s \rightarrow s} \leq 2^{1-2/s}(1 - \gamma)^{2d/s}.$$

For each fixed generator and guessed integer wrap, the full-environment affine map is a bijection preserving frame rank. The unique wrap is tested in integer coordinates *before* field reduction. Exactly one guess survives for each genuine step. Every surviving variant retains the original word’s weight; dividing by the number of guesses would be wrong and is not done. Contraction is applied to the complete conditional average before paying for individual variants in crude bounds.

4.6 Recursive precision and prefix estimates

High-level purpose. This block checks that scale changes, source changes, and successive-level differences leave enough statistical accuracy. An uncontrolled factor here would defeat the final moment estimate.

Status: conditionally verified through the averaging input.

R Lemmas 9.1–9.8 (**O** 7.1–7.8).

True sample means follow by conditioning on the retained midpoint and using the independent role channels. The homogeneous loss is comparable to the s th power of root discrepancy and contracts under nonnegative sums. Its relative-error inequality handles the signed residual expansion by bounding unsigned exact contributions against the nonnegative target.

The child error incurs scale power b/a for $b \geq a$ and $(b/a)^{s-1}$ for $b < a$, plus f/u exactly when the source is released. With $s = 8$ and the two half-log deductions in

$$p_{\text{acc}} = m/k_{\text{prec}} - \frac{1}{2} \log_2(1/a) - \frac{1}{2} \log_2(1/u) - a_* i_*,$$

the residual scale factors decay as $2^{-3|\log_2(b/a)|/8}$, and a released-source factor leaves $(f/u)^{-3/8}$ after taking the norm root. Fine omitted bins form a convergent geometric tail. The scale cutoff is chosen before using this tail in the induction.

At slack $d \leq m - r < 2d$, both pieces of a successive-level difference use the same walk and wrap choices. For its absolute raw bound, subtract the *same true stochastic term* from both pieces before taking the triangle inequality. Subtracting just the deterministic mean would leave large sampling noise and would not prove this bound. The mean-prefix telescoping reaches the last included child level, with coarse inactive children and omitted fine tails accounted for separately. The noise bound follows from conditional contraction and an adjustable fixed walk-length coefficient. The deterministic-mean premise is expressly discharged for the clean calculation, not assumed for arbitrary frozen-dependent functions.

4.7 Compression and clamping

High-level purpose. A bounded list must not destroy the statistical estimate or assume that approximation errors are independent of crowded-row events. This is the main nonlinear stability check.

Status: conditionally verified through the prefix estimates.

R Lemmas 10.2–10.5, Proposition 10.6, Lemma 10.7, Proposition 10.8.

The compression formula has baseline $u_0 \leq z' \leq 3u_0$ and radius $7(M_y - 2u_0)_+$. If $M_y \leq 2u_0$, the output is the baseline. If $M_y \geq 3u_0$, then

$$M_y + 3u_0 \leq 7(M_y - 2u_0),$$

so it is exactly the input, including for negative coordinates. Perturbing the input and baseline by δ perturbs u_0 by at most δ , z' by at most 3δ , and the radius by at most 21δ ; the stated constant 28 is sufficient. Fewer than the prescribed capacity have size strictly greater than u_0 . A buffered finite test can retain all possible exceptions without deciding exact equality.

Before first overflow, changes are bounded by $6 \sum \rho_i$. At a substantial true entry any such deletion requires a fixed lower bound on relative error, giving the stronger preloss charge. The stationary reference preserves the row rank when that rank exceeds the common flat region; otherwise it stays in that flat region. Entries in a relevant support exceed three times the active threshold and are untouched.

The relative-stability argument does not multiply an unbounded absolute true entry by a global Lipschitz constant. If at least k_{row} entries are huge, the gates close. If fewer are huge, common large sentinels preserve all relevant ranks and saturated values, reducing the comparison to bounded absolute errors. Removing the sentinels is justified because the original huge entries stay in the identity region.

False overflow is charged to at least half a capacity of erroneous light positions, at cost $(n_j \rho_j^s)^{-1}$. True-heavy rarity is conditioned separately at each endpoint, where the mean-error magnitude is deterministic. It supplies $(n_j \rho_j)^{-1/s}$ only to mean error. Noise is instead charged once using the disjoint first-overflow events. Thus no false independence of noise and overflow is invoked.

Tail-margin clamps preserve the needed row functions and witnesses for common exact-ID increments. They do *not* assert preservation of access-list presentations. The adaptive comparison uses two already-clamped calculations directly, with nonexpansiveness controlling their differing increments. This distinction removes a potentially invalid transfer of access lists from the unclamped proof.

4.8 Fingerprints, adaptive access, and the joint induction

High-level purpose. Pairwise hashing is insufficient for endpoints chosen adaptively using the hash. The proof must produce an independent comparison set and charge every departure from it.

Status: conditionally verified through the clean error reserve.

R Lemmas 11.1, 11.3–11.8, Proposition 11.9, Theorem 11.10.

The polynomial hash has collision bound

$$2(L_{\text{ID}} - 1)/q + 4/B + 4/q = O(B^{-1}), \quad q \asymp B^3.$$

The factor four correctly accounts for reducing two fair-bit chunks modulo a prime; uniform reduced residues are not assumed. Raw bit encodings are retained in the final enumeration. The four-visit catalytic equality test cancels arbitrary initial vector contents exactly. Repeated paths must ignore the accumulated result bit and incoming catalyst, which is part of the lower-operation induction.

For a fixed block, replacing *all* its equality tests by exact equality makes values, masks, port orders, and inverse accesses independent of its effective salt. Lower blocks are actual but cannot inspect this salt in their supplied coordinates. An ancestral transformation can read different physical master bits while replaying coordinates; this is not independence of the original master-bit labels. Each fixed variant permits the required change of variables. No uniform-law assertion is made for an adaptively chosen variant.

At the boundary before small-slack groups, the actual and clean lists coincide and have size $2^{O(h_*)}$. The envelope includes that entire boundary list and fixed-variant clean witness supports. Its size is $2^{O(h_*)}$, independent of the current salt. A valid actual endpoint outside it forces a child witness discrepancy of one. For products, the actual left correction's incoming cap removes the old-source sum even when that port has zero numerical weight. A bounded incoming count on the *clean witness* is neither needed nor claimed.

Only new increments are bucketed. Old representatives and their values are not merged; collisions add an increment to every old representative with that key. Consequently old array discrepancy enters with coefficient one, while the new collision charge is proportional to the small increment. The conditional hash bound is applied after fixing other coordinates, not after conditioning on an event depending on the current salt. The escape charge covers the latter dependence separately.

The simultaneous induction proves actual target accuracy, clean target accuracy, and their discrepancy. With

$$\lambda_m = 2^{-d_1 h_* + d_2(m-m_b)},$$

a child drop of at least d supplies $2^{-d_2 d}$. Choosing d_2 above the crude propagation exponent makes the child sum at most $\lambda_m/4$; choosing $d_1 > d_2 + 1$ and then sufficiently small ϵ makes the direct charge at most $\lambda_m/4$. The spare clean coefficient absorbs the result. Later enlargements of field size and top level do not affect the crude exponent: the component count, immediate descriptor growth per slack, and walk coefficient were already fixed.

4.9 Numerical operations and digit lifetimes

High-level purpose. Saving a full approximate number at each recursive level would exceed the claimed memory. The operand-place accounting must be verified for every primitive, not inferred from ordinary logspace arithmetic.

Status: verified.

R Section 12 and Appendix A (**O** Section 10).

If $|N_t - 2^t z| < 2$, the digits $z_0 = N_0$, $z_t = N_t - 2N_{t-1}$ telescope to z and lie in the stated redundant alphabet. A fixed modulus 2^q , $q \geq 4$, distinguishes the possible signed differences. The tail bound $6 \cdot 2^{-k}$ follows by a geometric sum. Gate addresses are regenerated from child-choice paths; a bounded-fan-in traversal saves bounded bits per gate, not one full address per gate.

For finite comparisons, the scaled prefix recurrence absorbs its sign once its magnitude exceeds 12; the remaining suffix has magnitude strictly less than 12. Min/max selection therefore compares finite truncations exactly, with the truncation error absorbed in the output approximation. Only constantly many low input places are needed for approximation residues.

Multiplication partitions pairs by the smaller digit index into dyadic bands. A band's truncation margin increases by a fixed fraction of its index, making its tail summable. Relevant diagonals and signed floor residues reduce each band's finite sum to bounded-fan-in arithmetic. The path cost accumulated over dyadic bands is polynomial in $\log(d+1)$ and bounded by $O(d)$; an operand in that band is early enough relative to the output place to pay it. This is a proof of the suspended-record bound, not just of computability of the product.

For reciprocal, the early prefix has length $\lfloor t/2 \rfloor + O(1)$. The exact identity

$$\frac{1}{Cx} = \frac{1}{Cx^0} - \frac{x - x^0}{C(x^0)^2} + \frac{(x - x^0)^2}{C(x^0)^2 x}$$

leaves a quadratic error below the fixed output margin. The prefix-dependent coefficients are rebuilt on their own leaves, and released before a late suffix query. Their words are not falsely classified as public constants. The inequality $t + 1 \leq 2(t - j) + O(1)$ pays for the early prefix at every input call that needs it.

Finite ranks use prescribed padded domains and a deterministic tie convention. Uniform perturbation of all entries controls the order statistic, including boundary ties. Balanced counting/summing circuits have the asserted depth in the descriptor length. Empty sums return zero; averages have a public nonempty denominator. Buffered tests place their fixed threshold between u and $2u$ using the positive floor. These structural choices are independent of a later numerical output clock.

4.10 Controller, inverse traversal, and totality

High-level purpose. This is the decisive algorithmic step: every active caller, inverse query, and inaccurate environment must fit simultaneously. A bound for only one frame would not prove logarithmic space.

Status: verified as an abstract uniform implementation.

R Theorem 13.1, Definitions 13.2–13.3, Lemmas 13.4–13.11 (**O** Section 11).

A graph state contains the cursor and the local explicit terminal chain. Opaque restoring queries and paired motions are edges whose internal workspaces are excluded from that graph. This does not make their workspace free: it becomes a new physical stack segment when called. Root boundaries are constant tags; absolute tuples are regenerated from relative recipes in shared scratch. The start row is not a root parameter of an incoming-search graph.

Every potentially invalid port is freshly tested at canonical entry. Historical flags in fabricated predecessor states are not trusted. Local edits erase bounded symbols at marked positions; therefore predecessor cases have bounded

descriptions. A paired-step predecessor first passes cursor-independent allocation and recipe guards, then uses the lower paired inverse, checks the returned token, and restores by the corresponding forward map. An unrelated value test is a separate edge. This prevents a hidden high-cost query while an extra predecessor address is live.

A deterministic outdegree-one component containing a success sink cannot contain a directed cycle or another sink. Its underlying component is a tree. Rotation and edge flip, with a sentinel for the isolated-root case, give a finite contour. Initiation marks count each row port once. Capped incoming access counts only up to the fixed cap and undoes a bounded number of mark jumps; it stores no contour-length counter or second endpoint.

For incoming score ranks, the quantized score belongs to the success state, not to immutable root parameters. During a contour it can be replaced by a partial score suffix and recovered at the unique sink. Digits are assembled from least to most significant, so a late operand query retains only $O(1 + (t - j)_+)$ suffix bits. Zero scores quantize to zero after clipping; absent positive entries are not lost by the support-covering class. The decreasing score loop terminates, including at isolated success states.

Actual recipes have a finite acyclic operation order: intrinsic validity and motion precede outer validity; comparison and matching precede updated bucket values; ranks and threshold tests precede shortlist lookup; incoming access follows pre-incoming values. Old-prefix calls double slack, and factor calls reduce accuracy level by at least slack. Excursions retain immediate path choices, perform only a pure local action at the remote endpoint, and discard the deeper spine before an earlier inverse query. Thus no unbalanced full path survives across a new tight endpoint query.

A fixed number of raw local-chain copies pays for predecessor probing. If a raw segment costs $a_0\Delta w + b_0\Delta t$ and $\Delta t \geq -a'\Delta w$, choose physical coefficients once so that

$$A \geq ka_0 + (D - kb_0)a' + c, \quad D \geq kb_0.$$

Then $k(a_0\Delta w + b_0\Delta t) + c \leq A\Delta w + D\Delta t$. Lower physical workspaces are not copied into upper raw graphs. Recursively applying a copy factor to the already enlarged physical bound would fail; the manuscript explicitly avoids it.

The allowance $w = k_1(1 + m) - (k_0 + s_0)d$ drops on same-prefix strata, previous-prefix calls, and lower-level calls when $k_0 > s_{\text{op}} + 1$ and $k_1 > 2(k_0 + s_{\text{op}} + 1)$. These drops dominate the concrete recipe costs. The potentials $Aw + Dt$ and $t + a'w$ bound total retained segments and all descendant clocks, respectively. Induction covers every admissible synthetic call as well as designated executions. Finite contours and finite local loops prove totality before configuration counting is used for time.

4.11 Final approximation and consequences

High-level purpose. An average-over-vertices moment bound does not automatically estimate one specified start. This final conversion also fixes the exact distribution the deterministic enumeration must reproduce.

Status: conditionally verified, using F1 and the imported averaging inputs.

R Section 14 and Appendix B (**O** Sections 12–13).

Keeping the single nonnegative summand at z and dividing by its retention probability gives

$$\mathbb{E}[|\bar{w}_z - w_z|^8 \mid X_f(z) = 1] \leq N(f/\pi_f)c_{\text{ind}}^8 2^{-8p_{\text{acc}}}.$$

The factor is N , not 1 or N^{-1} . The selected top precision pays for it. Hierarchy error, statistical error on at least 7/8 of the conditioned environments, and digit truncation sum to less than $3\Delta/8$. The upper median has the same guarantee. Enumerating all valid field matrices and all original fair-bit salt strings preserves multiplicities; enumerating just distinct reduced hash parameters would not.

The two-sided decision threshold has margins $11/24 < 1/2 < 13/24$, including $n = 0$. Two independent OR repetitions embed RL into BPL. The same total approximator yields total promise separation.

Variable precision uses one fixed exponent-one approximator on a virtual padded input of length $2q + \lceil \log_2(2n+4) \rceil + 4$. Rounding to denominator 2^{q+2} adds at most 2^{-q-3} ; padding error is below 2^{-q-4} . This proves the requested uniform bounds without letting construction constants depend on q .

For search, choosing a successor of maximum approximate continuation value loses at most 2δ per step. With $\delta \leq (n+2)^{-c}/(8T)$, the terminal value remains positive on the promised inputs and hence is accepting. A first pass verifies the resulting path without output; a second identical pass streams it. Off-promise failure cannot leak an invalid purported witness.

The compiler uses one total resource-capped randomized interpreter, one fixed deterministic approximation library, and virtual padding. Its tape and time caps apply even to malformed programs. For a promised source, the polynomial padding provides room for the encoded prefix and for the source’s time and traversed cells, preserving acceptance probability exactly. Fixed block encodings give a finite-control library with a syntactic tape bound; the compiler need not infer a semantic constant from a source program. The added address and callback words have explicitly bounded lengths. Counting the resulting total deterministic machine’s complete configurations yields the emitted polynomial time bound for the same execution.

4.12 Adversarial cases

High-level purpose. These cases test hypotheses that are easy to lose in a rewrite. They are attempts to falsify the stated argument, not evidence from random experiments.

Case	Check and result
Zero transition matrix; no active vertices	Corrections remain zero, no extra active copies appear, and the success vector already equals the exact probability. No division by active count is needed.
Many tiny edges	Small entries alone do not imply small row mass. The additional active-count bound is present and produces the factor $2^{-(h-1)l}$.
Deterministic forward chain	Full squaring preserves unit edges. The rewrite uses this as a warning, not as its density-reduction argument. Thresholded propagation plus copies supply the actual bound.
Column multiplier zero	Correction normalization is only lower-bounded on positive-multiplier support. Clipped denominators make the zero-output branch defined.
Negative and enormous raw estimates	Support gates work without positivity of estimates. Compression is Lipschitz; relative stability uses saturation and fewer-than-rank sentinels. These cases are explicitly covered.
Ties and exact threshold values	Ranks are Lipschitz under tied values. Port selection uses a fixed finite-margin test rather than equality of exact real numbers. Boundary uncertainty lies inside the permitted gap.
Empty or default-only row	Defaults have sufficient multiplicity in ranks to close gates before exposing positive values. Empty sums are zero and incoming ranks have zero padding.
Overfull column of zero-weight ports	The cap applies to exposed ports themselves; a value-only cap would fail the escape argument. The stated cap covers this case.
All or badly correlated hash collisions	Per-environment access and termination do not require hash accuracy. The statistical proof conditions on a salt-independent envelope and charges departures; it does not union-bound an adaptive set as fixed.
Isolated success state; cyclic artificial component	The sentinel handles the isolated state. A directed cycle cannot share an outdegree-one component with a success sink; only that sink component is toured.
Fabricated validity flags and missing ports	Lower calls start at canonical entry and repeat their own presence test. A flag alone never licenses a motion or digit query.
Rank-deficient frame; full-length frame	Rank-deficient frames are excluded by the probability law. A length- h frame would obstruct the stated transitivity proof; the actual hypothesis is $h_0 - 1 < h$.
Smallest input and precision	The $n + 2$ convention, fixed minimum padding of B , and unary $q \geq 1$ cover $n = 0$ and $q = 1$. None of the final error margins requires n to be large.
False compiler promise	The capped interpreter and fixed library are total. Language correctness is promised only when source bounds and error thresholds hold; this is the same restriction in both documents.

High-level purpose. The rewrite adds an illustrative numerical update. Checking it exactly prevents a pedagogical example from quietly asserting a different invariant.

Figure 1 has $C_{xy} = C_{yz} = 1/2$ and $w = (0, 1/4, 1)^T$, giving $p = (3/8, 3/4, 1)^T$. With $E = C$, the updated reward is $(1/8, 3/4, 1)^T$ and the sole residual edge has weight $1/4$ from x to z , so the start value is $1/8 + 1/4 = 3/8$. With only $E_{xy} = 1/4$, it is $1/16 + (1/4)(3/4) + 1/8 = 3/8$. The entries and stated preservation are correct. This is an exact check of that example, not a proof of the general construction.

4.13 Comparison with the original

High-level purpose. An exposition can preserve the headline theorem while losing an important promise or a technical interface. This comparison checks the formal results and their stated restrictions, separately from the independent proof review.

The source inventory contains, in each version, 53 lemmas, 7 theorems, 5 propositions, 2 corollaries, and 2 definitions: 69 formal environments. Both retain 67 proof environments and 7 remarks. Every original technical label is retained. These counts are a coverage check, not a correctness test. Comparing statement prose after separating mathematical expressions shows only terminology changes and an added clarification that atom assembly products and powers are entrywise; the main theorem statements were compared directly. The mathematical reading was separately checked throughout the proof blocks above.

No formal result is omitted. Machine conventions move from the original introduction into rewrite Section 3. Arithmetic proofs move to Appendix A while their interface remains in Section 12. The compiler moves to Appendix B. The catalytic comparison lemma is placed earlier. Historical context is moved to the end. The added overview, algorithms, example, and glossary are exposition; they do not add an expander-substitution theorem or a spectral sparsification guarantee.

The following restrictions are preserved: worst-case time on every coin sequence; fresh fair coins; confined input heads; counting traversed cells and the full stack; positive frame-rank and identifier-range restrictions; one retained identifier per conditioned channel; no independence assumption on estimated factors; total access on inaccurate environments; raw-salt multiplicity; fixed-source constants; unary precision; off-promise totality; and supplied compiler bounds. F1–F3 are notation damage introduced by the rewrite, not changes justified by the original.

Formal-result crosswalk

The columns give original and rewrite theorem numbers, then rewrite page. “V” means verified locally; “C” means conditionally verified through the listed imports or F1–F3. This is not a second principal verdict. Proof reasoning is in Sections 4.2–4.11. Definitions are checked for consistent use rather than treated as proved propositions.

O	R	Page	Result	Status
1.1	1.1	4	Class equality	C
1.2	1.2	4	Input-precision probability approximation	C
2.1	3.1	13	Configuration reduction	V
3.1	4.1	17	Rank-channel estimates	V
4.1	5.2	21	Fixed point and factorial truncation	V
4.2	5.3	22	Elementary pilot bounds	V
4.3	5.4	23	Detector bounds overloaded columns	V
4.4	5.5	24	Residual bounds and unsigned-term domination	V
4.5	5.6	26	Density and active-count reduction	V
4.7	5.8	27	Success contribution identity and shallow target	V
5.1	7.2	34	Dense atom identity	V
5.2	7.3	35	Fine product contribution	V
5.3	7.4	36	Correction atom identity, including zero multipliers	V
5.5	7.6	38	Support margins and exposed row classes	V
5.6	7.7	40	Port contract	V
5.7	7.8	41	Assembly comparisons	C
6.1	8.1	43	Uniform affine spectral gap	C
6.2	8.2	44	Conditional averaging	C
6.3	8.4	46	Validated walk variants	V
7.1	9.1	49	True sample means	C
7.2	9.2	50	Homogeneous loss	V
7.3	9.3	50	Scale and domain comparison	V
7.4	9.4	52	Fine omission	V
7.5	9.5	54	Immediate paths	V
7.6	9.6	54	Mean prefixes	V
7.7	9.7	55	Absolute differences	V
7.8	9.8	56	Conditional prefix bounds	C
8.1	10.2	59	Local compression properties	V
8.2	10.3	61	Preloss and stationary reference	V
8.3	10.4	62	Stability after first overflow	V

O	R	Page	Result	Status
8.4	10.5	63	Conditional rarity of a true heavy count	V
8.5	10.6	64	Integrated compression bound	V
8.6	10.7	66	Range clamping	V
8.7	10.8	67	Error reserve with exact endpoint grouping	C
9.1	11.1	68	Short fingerprints	C
9.9	11.3	70	Restoring equality test	V
9.2	11.4	72	Independence of the clean block	V
9.3	11.5	73	Envelope size and independence	C
9.4	11.6	74	Charging an escaped candidate	V
9.5	11.7	75	Only increments are aliased	V
9.6	11.8	76	Direct cost of collisions	C
9.7	11.9	77	Uniform block coupling	C
9.8	11.10	79	Actual-estimator error theorem; F1 applies	C
10.1	12.1	80	Exact streams from approximation residues	V
10.2	12.2	81	Evaluation from regenerated gate addresses	V
10.9	12.3	82	Uniform numerical interface	V
11.1	13.1	85	Realization of estimator interfaces	V
11.2	13.2	86	Definition: regenerable record	V
11.3	13.3	87	Definition: raw allowance certificate	V
11.4	13.4	88	Guarded local graph	V
11.5	13.5	91	Inverse contour	V
11.6	13.6	92	Capped incoming ports	V
11.7	13.7	93	Incoming score rank	V
11.8	13.8	94	Ordered terminal paths and excursions	V
11.9	13.9	95	Implemented endpoint comparison	V
11.10	13.10	95	Concrete recipe certificates	V
11.11	13.11	100	Parameter replay and salt visibility	V
12.1	14.1	104	Fixed inverse-polynomial accuracy	C
12.2	14.3	106	Conditioning on one start	V
12.3	14.4	109	Promise-class equality	C
12.4	14.5	110	Certified accepting computations	C
10.3	A.1	112	Finite comparisons and elementary operations	V
10.4	A.2	113	Multiplication	V
10.5	A.3	115	Reciprocal on a positive compact interval	V
10.6	A.4	117	Indexed finite ranks	V
10.7	A.5	118	Bounded sums and known scales	V
10.8	A.6	119	Finite buffered tests	V
13.1	B.1	119	Effective simulation with explicit bounds	C
13.2	B.2	121	Total resource-capped interpreter	V

5 Parameter, probability, and complexity audit

5.1 Compatible constants and finite ranges

High-level purpose. “Choose constants large enough” is valid only if the dependencies are acyclic. This block checks the order and the inequalities that make the main error sums converge.

The dimension, channel count, and $s = 8$ are fixed first. Correction thresholds determine the constant sweep count; routing determines a fixed radix. Row-gate rank precedes detector rank, overload threshold, and incoming cap. Precision/scale losses precede compression parameters and the walk coefficient. Crude error propagation precedes the fingerprint constants. Controller copy/allowance constants precede the top level and field-size exponent. None of the earlier spectral constants depends on that field exponent or the total bit length.

For the compression sums, $\rho_j = 2^{-c_1 d_j}$ and $n_j = 2^{c_2 d_j}$. With bias exponent b_{bias} , the sufficient choices

$$c_1 > b_{\text{bias}}, \quad c_2 > sc_1 + sb_{\text{bias}}$$

make false-overflow power decay as $2^{-(c_2 - sc_1 - sb_{\text{bias}})d_j}$ and the true-heavy mean sum decay, up to a fixed polynomial

in d_j , as

$$2^{-((c_2 - c_1)/s - b_{\text{bias}})d_j}.$$

The latter exponent is positive under the stronger preceding choice. Dyadic propagation is polynomial in slack, not exponential in the total number of stages. Increasing the fixed noise exponent controls the remaining polynomially weighted noise sum.

For fingerprints, $h_* = \lfloor \epsilon \log_2 B \rfloor$ is positive after fixed minimum padding, and $h_* \geq (\epsilon/2) \log_2 B$. Choose d_2 so that $\sum_{d \geq 1} 2^{-(d_2 - C)d} \leq 1/4$, then $d_1 > d_2 + 1$, then

$$1/(2s\epsilon) > C + d_1 + 2.$$

Thus $2^{Ch_*} B^{-1/(2s)} \leq \frac{1}{4} 2^{-d_1 h_*}$ after fixed padding. There are $O(B/h_*)$ salts, each of $O(\log B)$ bits: total $O(B)$. Constants may be extremely large; no practical efficiency claim is proved.

The descendant rate exponents are bounded by $hl_{\max} + c(m_{\text{top}} - m)$. Choosing the field last makes every identifier smaller than it, makes radix multiplication invertible, and ensures every retention box is nonempty. Trial division for a prime of size $2^{O(B)}$ is uniform in $O(B)$ space and $2^{O(B)}$ time. It is not being claimed polynomial in the field's bit length.

5.2 Probability and final errors

High-level purpose. The final majority must refer to the same finite distribution that is enumerated, and every loss must fit within the requested tolerance.

All probability here is over independent rank channels conditioned on frame success and independent raw fair salt strings. A conditional slice additionally fixes specified retention events and any frozen effective bits. Those retention events have strictly positive probability. Role-channel independence is used only at the true identities. Conditional pair bounds replace stronger independence assumptions in gate and heavy-count estimates.

With $r = d \lceil \log_2(n+2) \rceil$ and $\Delta = 2^{-r}$, choose

$$l_{\max} = \left\lceil \frac{c_0 B + r + 3}{h - 1} \right\rceil, \quad p_{\text{acc}} \geq r + 3 + \frac{c_N B + \lceil \log_2 C_{\text{tail}} \rceil + 3}{8}.$$

The hierarchy error is at most $\Delta/8$. The conditional statistical failure probability at threshold $\Delta/8$ is bounded by

$$C_{\text{tail}} 2^{c_N B - 8p_{\text{acc}} + 8r + 24} \leq 1/8.$$

Numerical truncation through place $r + 6$ adds at most $62^{-r-6} < \Delta/8$. More than three quarters of the conditional encodings therefore lie within $3\Delta/8$ of the answer. No union bound over all vertices is needed. The conditional median inherits this guarantee even if every bad estimate is adversarially placed.

5.3 Simultaneous space, time, and output

High-level purpose. Finite enumeration and recomputation can still violate logarithmic space if each caller keeps its own addresses or numerical words. This ledger records what remains live.

Live object	Bound and justification
Master environment and all salts	$O(B)$ once; fixed-dimensional field matrices plus the block-salt bound.
Cursor, catalyst, shared scratch, top parameters	$O(B)$ once for actual top clock and level $O(B)$. The catalyst is counted ordinary memory.
All nested structural/numerical records and saved local probe copies	Sum of potential drops is $O(m_{\text{top}} + t_{\text{top}} + 1) = O(B)$. Delimiters, pending bits, and inverse tokens are included.
Absolute child tuples and temporary address copies	Regenerated only in shared scratch; none survives another recursive graph query.
Outer seed index, median counts, search bounds, numerator, and digit index	A fixed number of $O(B)$ -bit words, live once throughout a query and explicitly counted.
Input heads	Physically confined; virtual or off-data positions are represented in the already counted address words.

Live object	Bound and justification
Output	Written only after the fixed-accuracy median is computed; never readable as work memory. Search uses two passes to verify before streaming.

For a fixed input, the total deterministic query has $2^{O(B)}$ complete configurations, including tape heads and finite control. Totality excludes repetition before halt. Each query therefore takes $2^{O(B)}$ steps. Multiplying by $2^{O(B)}$ environments and polynomially many digit/counting passes still gives $2^{O(B)} = (n+2)^{O(1)}$. This argument is valid because there are no unit-cost external estimator oracles in the final machine.

For the compiler, let d_{code} be description length and $b_n = \lceil \log_2(n+2) \rceil$. The padding coefficient

$$p = 1000(d_{\text{code}} + 2)^4(a + b + 2), \quad N_{\text{pad}} = 2^{p b_n}$$

is fixed by the compiler input. The interpreter’s permitted strip exceeds the source’s promised traversed range, and its time cap exceeds $(n+2)^a$. The virtual-input callback stores only a fixed number of address and boundary words of length $O(\log N_{\text{pad}}) = O(\log(n+2))$ for this compiled machine. Tape-group widths and finite-control/state counts give explicit numerical output bounds. This is source-specific effective compilation, not a uniform small constant independent of the source description.

6 Citation and assumption audit

High-level purpose. The conditional verdict must expose exactly which outside mathematics is being granted. A matching citation title is not sufficient; the needed form and its application are recorded here.

- I1. **Shalom (1999), Theorem 2.1 and Corollary 2.3, pp. 152, 154.** Exact statement and projection application checked in the primary PDF. The relative constant is $1/10$ for the eight indicated generators of $\text{SL}_2(\mathbb{Z}) \ltimes \mathbb{Z}^2$; they occur in the manuscript’s generator set. The translation-fixed subspace and its orthogonal complement are invariant under the pair subgroup, so the complement application is legitimate. The published theorem is granted, not re-proved here.
- I2. **Shalom (1999), Theorem 2.6, p. 156.** Exact statement checked: for dimension at least three the elementary generators have a positive quantitative Kazhdan constant $(20\nu_h)^{-1}$. Adjoining inverse generators is harmless. Only a dimension-dependent constant is required. Its bounded-generation input is part of the granted theorem, not an independently rechecked proof in this audit.
- I3. **Riesz (1927), Theorems II and II’, p. 472.** The finite weighted-norm interpolation statement was checked against the primary source. Its diagonal between L_2 and L_∞ gives the manuscript’s displayed L_s estimate for $2 \leq s < \infty$. The supplemental Thorin citation is unnecessary to this finite real-valued application and was not independently audited.
- I4. **Bertrand’s postulate.** Granted standard import: for each integer $n > 1$ there is a prime between n and $2n$. It supplies both the environment prime and the much smaller hash prime. It was not separately sourced or re-proved here. Trial-division space and time accounting, and positivity of the finite ranges, were checked directly.

Cook–McKenzie (1987), Proposition 1, was checked in the primary source for the forest contour claim. The manuscript also proves the needed rotation/flip statement, including the added isolated-root sentinel, so its use does not leave an additional unproved load-bearing import. Markov, Chebyshev, the finite-dimensional rank and determinant manipulations, finite geometric sums, triangle/Hölder/Jensen bounds used in the error calculations, polynomial root counting, and deterministic configuration counting were checked directly at their uses. No conjecture, computational experiment, or unproved derandomization assumption is being granted.

Primary external sources consulted

- Shalom: https://www.numdam.org/item/PMIHES_1999__90__145_0.pdf
- Riesz: https://archive.ymsc.tsinghua.edu.cn/pacm_download/117/5399-11511_2007_Article_BF02564121.pdf
- Cook–McKenzie: https://www.cs.toronto.edu/~sacook/homepage/cook_mckenzie.pdf

The other bibliography entries are historical or comparative, not premises of the checked construction. Their original proofs and all of their precise historical attributions were **not independently checked**. This includes the overview’s comparisons with cut sparsification, zig-zag expansion, Reingold’s connectivity algorithm, and Laplacian solvers, as well as the related-work seed-length and time-space bounds. The audit does check that these comparisons are not substituted for the manuscript’s own conditional sampler or controller proof. It does not certify priority, novelty, completeness of the literature survey, or every bibliographic claim.

Reproducible document identifiers

Immutable PDF links:

- [Pinned rewrite PDF](#), with adjacent `main.tex` used for source comparison.
- [Pinned original PDF](#), with its corresponding `build` source files.

SHA-256 values:

Rewrite PDF

332d364f809e522863089c77757e30ffb2701e9fd3d50bd04caaf8c800bbf8d5

Original PDF

15030fab23d31a4d91bd944100a547c3e9eff6da8330a4fb94ea999f071fadfc

Attached verifier

0eb1b3aad6ab6c76d08c9688002d1ddac9ded2c53594861790d3d1bc85d55a3

Both PDFs were text-extracted; the three defective passages and the controller allowance page were also visually inspected to confirm the symbols are present in the PDFs and are not extraction artifacts. Source environment/label inventories and direct statement comparisons support the crosswalk. No repository tests or numerical experiments were used as a substitute for proof.

7 Minimal repair path

High-level purpose. These edits address the demonstrated defects only. They do not propose a new argument to cover an unproved theorem.

1. In Eq. (11.14), replace $x \in \mathbf{v}_l$ by $x \in V_l$.
2. After Eq. (7.16), replace J th by j_{det} th.
3. On p. 33, replace the reference to R_i by the value $\mathbf{r}_{i,x}^{\text{gate}}$ from Eq. (5.7), with its current row and sweep arguments.
4. Optionally rename the locally defined left relation $\mathcal{P}_{l_{\max}}^{\text{ac}}$ to avoid implying hierarchy-depth indexing.

The first three corrected readings have been checked against the original and the rewrite’s own defining formulas; they do not require an altered mathematical hypothesis. A future revised PDF should be checked for these precise substitutions and for accidental collateral changes. Removing the conditional status of this report would additionally require the reviewer to accept the granted external theorems as established inputs under their usual conventions, rather than treating their proofs as within the audit’s scope. No missing lemma or counterexample-driven change to the proof strategy was identified.

8 Residual uncertainty

The residual conditions are explicit: I1–I4 are imported mathematics; the literal text contains F1–F3; historical/comparative citations are not fully audited. The complete finite-control transition table of the proposed estimator/compiler was not generated or run, and no machine-checked formalization was supplied or constructed. The implementation verdict is based on the manuscript’s uniform recipes, dependency order, and full-stack bounds, whose arguments were reviewed above. This report therefore should not be described as experimental execution or formal-kernel verification of the claimed algorithm.

I found no unresolved load-bearing mathematical step within that conditional scope. This is a statement about this audit, not a guarantee that another adversarial review cannot discover an error. The repository comparison establishes preservation of the formal claims apart from the stated notation defects; it does not turn agreement between two documents into independent evidence for the theorem.

Notation glossary

High-level purpose. This glossary records the report’s aliases and the main mathematical objects so that its calculations can be compared directly with the rewrite.

Symbol	Meaning
R, O	Rewrite and original, at the pinned revisions above.
$n, \mathcal{M}, p_{\mathcal{M}}(x)$	Input length, fixed randomized machine, and its acceptance probability.
B, N	Report aliases for the rewrite’s $b_{\text{mem}} = \Theta(\log(n + 2))$ and $n_v = 2^{c_N B}$ stage-size bound.
$V_l, l_{\max}, h$	Stage vertex set, final hierarchy depth, and fixed sampling dimension/rate exponent.
S, e, p₀	Base forward transition, final acceptance indicator, and exact continuation-probability vector.
C, E, D, w	Stage transition, cumulative correction, residual transition, and success contribution.
g, P_l^c, B_l	Frozen column multiplier, cumulative weighted projection, and cumulative reward matrix.
f, a, u, π_a	Stage density, entry scale, source rate, and actual retention probability at rate a .
$m, r, d, p_{\text{acc}}, i_*$	Parent accuracy level, child level (or locally final bit exponent), slack (or locally fixed requested exponent), effective error exponent, and dependency rank. Local uses are specified where they occur.
s, h_*, λ_m	Moment power 8, salt-block width, and actual–clean discrepancy allowance.
ρ_j, n_j	Compression floor and capacity at group j .
w, w', t, j	Parent/child controller allowances and requested output/operand digit clocks in the memory calculation.
C, c, A, D	Fixed constants local to the displayed estimates; C is also the fixed normalization in the reciprocal identity. They are not additional matrices.
$\Delta, N_{\text{pad}}, d_{\text{code}}$	Final target tolerance, virtual compiler input length, and source-code length.